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Data handling & Visualization | 192225102 | Slot-C

Activity-6

a) Hidden Flows :-

Using Conservation of flow (inflow = outflow):

1.) Apply coupon → No order.
= 2658 - 824 = 1834 sessions

2.) Refferal code → Menu page = 970 sessions.

3.) Home Page → Exit site = 409 sessions.

b) Calculations :-

1.) Conversion rate : $\frac{1566}{6720} \times 100 = 23.30\%$

2.) Largest flow into No order / Exit site : 2,160 sessions

3.) Home Page → Add item to basket =
$\frac{949}{2591} \times 100 = 36.63\%$

c) Four design decisions :-

1.) Arrang nodes in journey order

2.) Use consistent colours for sources and outcomes.

3.) Label nodes and flows clearly

4.) Highlight flows ending in order paid.

(d.) Node heights:-

formula: Height = Sessions ÷ 300.

* menu page: 1774 / 300 = 5.91cm
* offers page: 2355 / 300 = 7.85cm
* Home page: 2591 / 300 = 8.64cm

(e.) R-code:-

```
library (networkD3)
nodes ← data.frame( name = c("source", "Landing",
                             "Action", "Outcome"))

links ← data.frame (
    source = c(0, 1, 2),
    target = c(1, 2, 3),
    value = c(2621, 2355, 1566)
)
SankeyNetwork (
 links = links, Nodes = nodes,
Source = "source", Target = "target",
value = "value", NodeID = "name"
)
```

(f.) Two limitations:-

1.) Too many crossing flows reduce readability.
use a grouped bar chart.

2.) Small flows are hard to compare;
use a flow table.